Quacklytics

AI Instruments for Healthcare & Pharma

AWS-Native Platform — Illuminate & Clinic

Development Plan & Implementation Guide

Version 3.1 — October 20, 2026

"From drug name to boardroom-ready pharma infographic in under 10 minutes — zero manual prompt engineering."

Author: Molly Maskrey, Lead Data Scientist
Organization: Quacklytics
Infrastructure: Personal AWS account (root: mollymaskrey, IAM user: aiengineer)
Pipeline: API Gateway → Lambda → Claude API → gpt-image-2 → S3 → CloudFront

Internal R&D document. AWS-native serverless pipeline.
Anthropic Claude + OpenAI gpt-image-2.
Pharmaceutical and clinical data sourced from public FDA, NIH, CMS, and Merck APIs.

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1 Quick Reference — Critical Identifiers

> **Note**
>
> **Keep this section current.** These are the identifiers you will need every time you update, redeploy, or debug any part of the platform. Do not hunt for these in the narrative sections.

1.1 AWS Account & IAM

Resource	Value
AWS Account ID	984270705076
Root account	mollymaskrey
IAM user	aiengineer
IAM group	QuacklyticsAutoViz
IAM execution role	illuminate-api-handler-role-wfc5mfjl
Self-invoke policy	illuminate-api-handler-self-invoke-policy
Region	us-east-1
GitHub repo	mollymaskrey/quacklytics-autoviz

1.2 API Gateway Endpoints

Product	API ID	Invoke URL
Illuminate	ozxodj0812	`https://ozxodj0812.execute-api.us-east-1.amazonaws.com/dev`
Clinic	wyg1dtb7pj	`https://wyg1dtb7pj.execute-api.us-east-1.amazonaws.com/dev`

1.3 Lambda Functions

Function	Product	Handler
illuminate-api-handler	Illuminate	api_handler.lambda_handler
clinic-api-handler	Clinic	api_handler.lambda_handler
Both: Python 3.12, 3008MB, 5min timeout, layer: illuminate-dependencies v1		

1.4 CloudFront Distributions

Distribution Domain	Serves	Origin
d265he30e9rch9.cloudfront.net	quacklytics.org (landing)	quacklytics.org S3 bucket
duvaokvqrh5ct.cloudfront.net	illuminate.quacklytics.org	illuminate-quacklytics-org S3 bucket
d3hu7iaz3hezk9.cloudfront.net	clinic.quacklytics.org	clinic-quacklytics-org S3 bucket
dapcp3jomj7rx.cloudfront.net	Infographic PNG delivery	autoviz-artifacts S3 bucket

> **Note**
>
> **d265he30e9rch9 is on the Pro plan ($15/month).** The other three distributions are on the Free plan. CloudFront free plan limit is 3 distributions per account — the quacklytics-website distribution was upgraded to Pro to unlock the 4th slot for Clinic.

1.5 S3 Buckets

Bucket	Purpose
quacklytics.org	Portfolio landing page index.html
illuminate-quacklytics-org	Illuminate app index.html
clinic-quacklytics-org	Clinic app index.html
autoviz-artifacts	All generated PNGs + cached source JSON + style files

1.6 DynamoDB Tables

Table	Keys	Purpose
IlluminateIndex	PK: drug_name / SK: indication	Illuminate source + infographic cache
IlluminateJobs	PK: job_id	Illuminate async job tracking
ClinicIndex	PK: condition / SK: drug	Clinic source + visual cache
ClinicJobs	PK: job_id	Clinic async job tracking

> **Note**
>
> **DynamoDB reserved words:** `condition`, `language`, and `status` are reserved words in DynamoDB. Any scan or query that projects these fields must use `ExpressionAttributeNames` with placeholder aliases (e.g. `#cond` for `condition`, `#lang` for `language`). Failure to do so causes silent scan errors returning empty results.

1.7 Secrets Manager

Secret ID	Contains
/autoviz/openai_api_key	OpenAI API key for gpt-image-2
/autoviz/anthropic_api_key	Anthropic API key for Claude

1.8 ACM Certificate

Property	Value
Cert ID	d62a331f-6b3e-4cd3-8377-a5cd773a9da3
Covers	*.quacklytics.org and quacklytics.org
Region	us-east-1 (required for CloudFront)
Status	Issued

1.9 S3 Style Files

S3 Key	Purpose
style/quacklytics_dark_v1.txt	Master dark bioluminescent style (Illuminate)
style/templates/mechanism_view_v1.txt	Illuminate MechanismView template
style/templates/clinical_brief_v1.txt	Illuminate ClinicalBrief template
style/templates/disease_map_v1.txt	Illuminate DiseaseMap template
style/clinic/clinic_base_v1.txt	Clinic warm clinical style

1.10 Live URLs

URL	Status
https://quacklytics.org	Portfolio landing page — LIVE
https://illuminate.quacklytics.org	Illuminate app — LIVE
https://clinic.quacklytics.org	Clinic app — LIVE

1.11 Deployment Commands

Redeploy Illuminate Lambda:

```
cd ~/projects/AUTOVIZ/shared
rm -rf build_zip illuminate-handler-new.zip && mkdir build_zip
cp api_handler.py job_store.py duckdb_merger.py prompt_engineer.py \
   research_agent.py storage_agent.py synthesis_agent.py build_zip/
cp -r source_clients build_zip/source_clients
find build_zip -name "__pycache__" -type d -exec rm -rf {} +
cd build_zip && zip -r ../illuminate-handler-new.zip . && cd ..
# Upload illuminate-handler-new.zip to illuminate-api-handler Lambda
```

Redeploy Clinic Lambda:

```
cd /Users/mollymaskrey/projects/AUTOVIZ/clinic/lambdas
zip -r ~/Desktop/clinic-handler.zip .
# Upload clinic-handler.zip to clinic-api-handler Lambda
```

Invalidate CloudFront cache after index.html update: CloudFront → select distribution → Invalidations → Create invalidation → path: `/*`

2 Project Overview

2.1 What Illuminate Does

Quacklytics Illuminate is a fully automated, AWS-native pipeline that accepts a drug or disease name, input type, and audience level as input and returns a boardroom-ready pharmaceutical infographic as a CloudFront-delivered PNG URL — in under 10 minutes, with zero manual prompt engineering required.

The pipeline orchestrates nine pharmaceutical data sources, a Claude synthesis agent, a prompt engineering agent, and gpt-image-2 image generation across a serverless Lambda architecture, fronted by API Gateway and delivered to a live web UI at illuminate.quacklytics.org. **DuckDB** runs in-process inside the Research Agent as the multi-source data wrangling engine. All source

data and artifacts are stored permanently in S3, indexed via DynamoDB for fast lookup, with a 90-day staleness threshold governing automatic refresh.

2.2 Drug vs. Disease Mode

	Drug Mode	**Disease Mode**
Input	Generic or brand drug name	Disease or indication name
Sources	DailyMed, ClinicalTrials, openFDA, PubMed, RxNorm, CMS ASP (Part B) or CMS Part D (auto-routed), DrugBank (optional)	ClinicalTrials, PubMed, Merck Manual
Templates	MechanismView, ClinicalBrief	DiseaseMap
Pricing	Part B: J-code + ASP + ASP+6% Part D: total spending, claims, avg cost/beneficiary	Not applicable

2.3 Architecture Summary

Browser → **API Gateway** → **Lambda** → **Pipeline** → **gpt-image-2** ↓ **PNG** → **Storage Agent** → **S3** → **CloudFront** → **Browser**

2.4 Named for the Duck

Illuminate is part of the Quacklytics platform — named for DuckDB, the columnar query engine at the heart of the analytics architecture. *Quack* is the DuckDB CLI prompt itself. **Analytics powered by the duck.**

2.5 Pipeline Architecture

2.6 Strategic Context

This pipeline is developed on Molly Maskrey's personal AWS account outside IT-managed infrastructure. The demo target is Ed's return — a live drug-name-in, infographic-URL-out demonstration at illuminate.quacklytics.org. Authentication (Cognito) and Next.js/Amplify migration deferred until after Ed and Darren review.

3 Image Resolution & Pricing Tiers

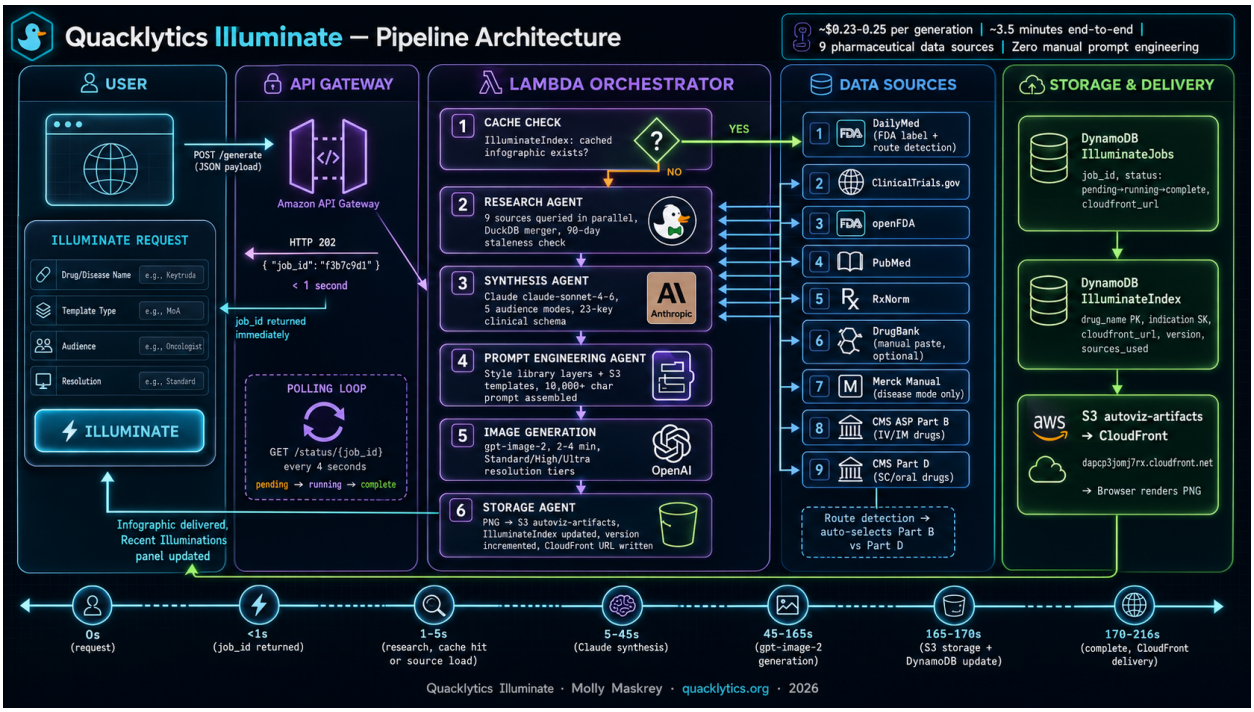


Figure 1: Quacklytics Illuminate pipeline architecture. Shows the full data flow across five swim lanes: User, API Gateway, Lambda Orchestrator, Data Sources, and Storage & Delivery. Bottom strip shows actual timing.

Tier	Resolution	Status	Est. Cost	Best For
Standard	2048×1152	Stable	~$0.19	Internal review, rapid iteration
High	2560×1440	Stable	~$0.25	Client presentations, boardroom demo
Ultra	3840×2160	Experimental	~$0.40	Print, large format display

Note

High (2560×1440) is the recommended default. Actual observed cost is $0.23–0.25 per generation.

4 Domain, Frontend & Infrastructure Architecture

4.1 Domain & Subdomain Strategy

URL	Purpose	Status
`quacklytics.org`	Portfolio landing page	LIVE ✅
`illuminate.quacklytics.org`	Illuminate app	LIVE ✅
`clinic.quacklytics.org`	Clinic app	LIVE ✅
`canvas.quacklytics.org`	Canvas app	Future

4.2 ACM Wildcard Certificate

Wildcard cert `*.quacklytics.org` issued in ACM us-east-1 covers all current and future subdomains. See Quick Reference for cert ID.

> ⚠ **Note**
>
> ACM certs must be in us-east-1 for CloudFront. The wildcard cert covers both `*.quacklytics.org` and bare `quacklytics.org` — the bare domain must be added explicitly even with a wildcard. `AmazonRoute53FullAccess` must be attached to the `QuacklyticsAutoViz` IAM group or Route 53 operations will fail.

4.3 CloudFront Distributions

See Quick Reference Section 1.4 for the full distribution table.

> ⚠ **Note**
>
> CloudFront free plan limit is 3 distributions. The `quacklytics-website` distribution (d265he30e9rch9) was upgraded to Pro plan ($15/month) to unlock the 4th slot for Clinic. quantum.mollymaskrey.com distribution was deleted to free a slot earlier. Always invalidate with `/*` after uploading new `index.html`. Do NOT invalidate `dapcp3jomj7rx` — PNG artifacts are immutable.

4.4 Portfolio Landing Page

The `quacklytics.org` landing page is a four-product portfolio hub in the 3D glassy bioluminescent style with per-product accent colors: Illuminate (cyan, Live), Canvas (purple, Coming Soon), Clinic (green, Coming Soon), Resolve (amber, Coming Soon). Contact: `mollymaskrey@gmail.com`.

5 Storage & Index Architecture

5.1 Design Philosophy

All source data and infographics stored permanently in S3. DynamoDB index checked first on every request. 90-day staleness threshold for source data refresh.

5.2 S3 — autoviz-artifacts Public Access

Note

Block Public Access OFF. Bucket policy allows `s3:GetObject` from any principal:

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Sid": "PublicReadGetObject",
    "Effect": "Allow",
    "Principal": "*",
    "Action": "s3:GetObject",
    "Resource": "arn:aws:s3:::autoviz-artifacts/*"
  }]
}
```

Same policy applies to `illuminate-quacklytics-org` and `clinic-quacklytics-org` buckets. Static website hosting must be enabled on app buckets with `index.html` as default document.

5.3 DynamoDB Tables — Illuminate

Table: `IlluminateIndex`. PK: `drug_name`. SK: `indication`.

Attribute	Type	Notes
drug_name	String (PK)	Lowercased slug
indication	String (SK)	"drug" or "disease"
last_refreshed	String	ISO 8601 UTC — source data age
last_generated	String	ISO 8601 UTC — last infographic
cloudfront_url	String	Must use dapcp3jomj7rx domain
s3_key	String	S3 key of most recent infographic
s3_source_keys	Map	Source → S3 key
sources_used	List	Sources that returned data
template_type	String	Last template used
audience	String	Last audience mode used
resolution	String	Last resolution used
version	Number	Increments on each generation

Table: `IlluminateJobs`. PK: `job_id`.

Attribute	Type	Notes
job_id	String (PK)	uuid4().hex
status	String	pending / running / complete / failed
params	Map	Validated request params
cloudfront_url	String	Present once status = complete
error	String	Present if status = failed
created_at	String	ISO 8601 UTC
updated_at	String	ISO 8601 UTC

5.4 DynamoDB Tables — Clinic

Table: `ClinicIndex`. PK: `condition`. SK: `drug`.

> **Note**
>
> **Reserved words:** `condition` and `language` are DynamoDB reserved words. All scan/query operations must use `ExpressionAttributeNames` with aliases (`#cond`, `#lang`). Failure causes silent empty results — this was the root cause of the Recent Clinics panel showing nothing after generation.

Attribute	Type	Notes
condition	String (PK)	Lowercased slug — reserved word, use #cond alias
drug	String (SK)	Lowercased slug
last_refreshed	String	ISO 8601 UTC
last_generated	String	ISO 8601 UTC
cloudfront_url	String	dapcp3jomj7rx domain
s3_key	String	S3 key of most recent visual
s3_source_keys	Map	Source → S3 key
persona	String	Last persona used
language	String	Last language — reserved word, use #lang alias
resolution	String	Last resolution used
version	Number	Increments on each generation

Table: `ClinicJobs`. PK: `job_id`. Same schema as `IlluminateJobs`.

6 Development Priorities & Task Order

Phase	Focus	Status
1	DALL-E image generation loop	COMPLETE ✅
2–3	All 9 data sources, Part B/D routing	COMPLETE ✅
3.5	DuckDB merger module	COMPLETE ✅
4a–4d	Research, Synthesis, Prompt Eng., E2E test	COMPLETE ✅
5a–5f	Storage Agent, API Handler, API Gateway, Lambda layer	COMPLETE ✅
6a	Full async pipeline	COMPLETE ✅
7	URL architecture, wildcard cert, subdomain routing	COMPLETE ✅
8	Quacklytics Clinic pipeline, Lambda, API Gateway	COMPLETE ✅
8b	Clinic subdomain DNS + CloudFront + frontend	COMPLETE ✅
— ILLUMINATE DEMO READY — CLINIC LIVE —		
6b	Authentication — Cognito User Pool	Deferred
6c	Amplify web UI	Deferred

7 Phases 1–4: Pipeline Core — ALL COMPLETE ✅

✅ Milestone

All nine data sources, DuckDB merger, Research Agent (Part B/D routing), Claude Synthesis Agent (five audience modes), and Prompt Engineering Agent complete and validated end-to-end.

7.1 Quick Reference — Nine Sources

#	Source	Mode	File
1	DailyMed FDA Label (+ route detection)	Drug	dailymed_client.py
2	ClinicalTrials.gov	Drug+Disease	clinicaltrials_client.py
3	openFDA	Drug	openfda_client.py
4	PubMed	Drug+Disease	pubmed_client.py
5	RxNorm	Drug	rxnorm_client.py
6	DrugBank (paste)	Drug, optional	drugbank_client.py
7	Merck Manual	Disease + Clinic condition	merck_client.py
8	CMS ASP (Part B)	Drug, IV/IM	cms_asp_client.py
9	CMS Part D	Drug, SC/Oral	cms_partd_client.py

8 Phase 5: Storage Agent, API Handler & API Gateway — COMPLETE ✅

✅ Milestone

Full backend wiring complete and live. API Gateway deployed to dev stage. Lambda packaged with dependency layer and correct IAM permissions.

8.1 API Gateway — Illuminate

API Name: `illuminate-api` (ozxodj0812), stage: dev. Resources: POST /generate, GET /recent, GET /status/{job_id} — all Lambda proxy integration to `illuminate-api-handler`.

8.2 Lambda Packaging — Illuminate

Handler: `api_handler.lambda_handler`. See Quick Reference for build and deploy commands.

IAM execution role `illuminate-api-handler-role-wfc5mfjl`: `AmazonS3FullAccess`, `AmazonDynamoDBFullA` `SecretsManagerReadWrite`, inline policy `illuminate-api-handler-self-invoke-policy` granting `lambda:InvokeFunction` on both Lambda ARNs.

9 Phase 6: Full Async Pipeline — COMPLETE ✅

Milestone

Demo-ready as of 2026-07-01. Four infographics confirmed live. Cost: $0.23–0.25 per generation. Time: 197–217 seconds.

POST /generate creates a pending job, self-invokes async (`InvocationType='Event'`), returns HTTP 202 with job_id in under 1 second. Background Lambda runs full pipeline, transitions pending → running → complete. Browser polls every 4 seconds.

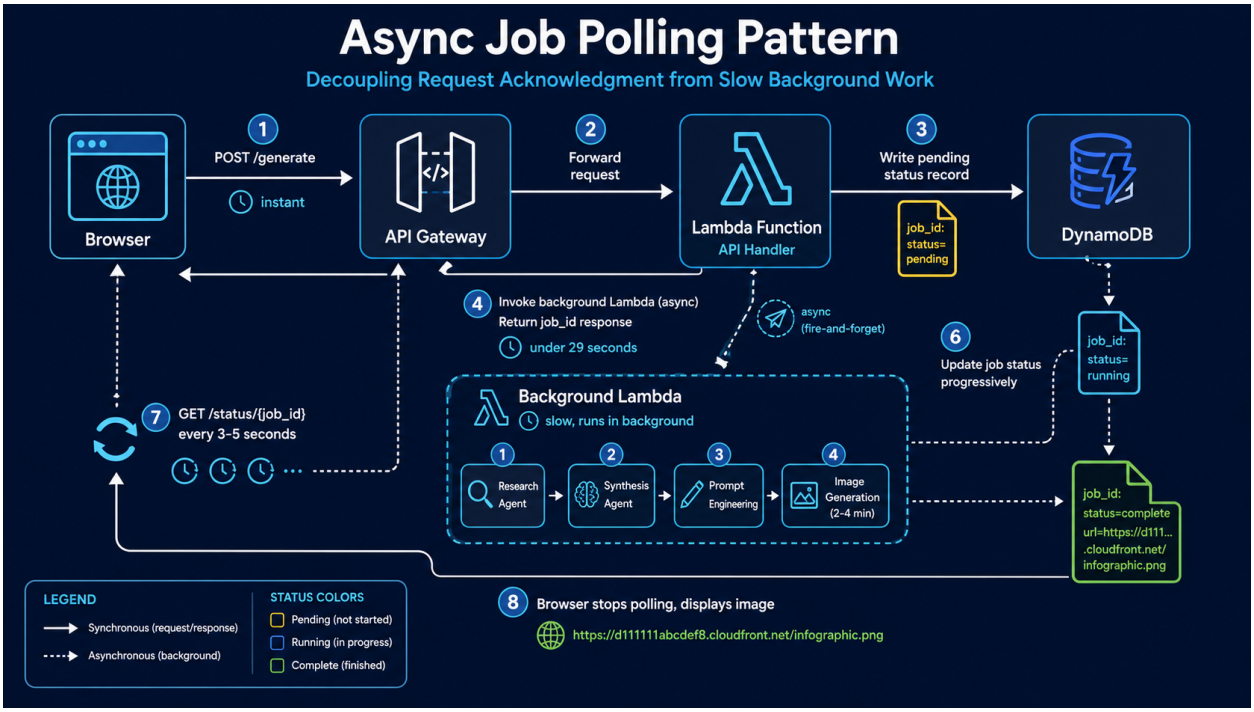


Figure 2: Async job polling sequence as implemented.

Tasks 6b (Cognito) and 6c (Amplify) deferred until after Ed/Darren review.

10 Quacklytics Clinic — LIVE

10.1 Product Concept

A point-of-care patient education instrument. Clinician enters condition, drug, patient persona, and language. Pipeline produces a patient-facing visual for display on a monitor in the exam room — replacing printed handouts that patients discard.

Key differentiation from Illuminate:

- Input: condition + drug pair, not a single drug/disease name
- Output: patient-facing education, not boardroom analytics
- Six patient personas replace five audience modes
- Eight languages — synthesis agent translates all patient-facing content
- Visual style: warm and accessible (clinic_base_v1.txt)
- Buyer: health systems and practices, not pharma analytics

10.2 Patient Personas & Languages

Persona	Description
general_adult	Baseline plain English, balanced text/visual
older_adult	Larger text, simpler language, daily impact focus
child_parent	Parent of child patient, hopeful tone, minimal jargon
non_english	Maximum visual-to-text ratio, universal symbols
tech_professional	Mechanism, statistics, clinical framing
clinician	Peer-level clinical reference card, full terminology

Languages (48 total combinations): English, Spanish, Mandarin, French, Portuguese, Arabic, Hindi, Vietnamese — all personas available in all languages.

10.3 Clinic Content Schema (15 Keys)

condition_name, condition_plain_summary, what_causes_it, how_it_affects_you, drug_name, drug_plain_summary, how_it_works, how_its_given, what_to_expect, most_important_side_effects, what_to_tell_your_doctor, how_well_it_works, questions_to_ask, key_takeaway, follow_up_note.

10.4 Architecture Comparison

Component	Illuminate	Clinic
Input	drug_name + input_type	condition + drug
Audiences/Personas	5 audience modes	6 patient personas
Language	English only	8 languages
Research	Single source set	Condition + drug sources merged
Synthesis schema	23 drug / 11 disease keys	15 patient education keys
Visual style	Dark bioluminescent	Warm clinical
DynamoDB	IlluminateIndex / IlluminateJobs	ClinicIndex / ClinicJobs
S3 prefix	infographics/	clinic/
Lambda	illuminate-api-handler	clinic-api-handler
API Gateway	illuminate-api (ozxodj0812)	clinic-api (wyg1dtb7pj)

10.5 Clinic Pipeline Architecture

10.6 Confirmed Generations

Condition	Drug	Persona / Language	Time
Type 2 diabetes	Tirzepatide	General Adult / English	3m 12s
Atrial fibrillation	Rivaroxaban	Older Adult / Spanish	2m 57s
Obesity	Tirzepatide	Tech Professional / English	~3m

10.7 Known Issues & Fixes Applied

- **DynamoDB reserved words:** `condition` and `language` are reserved. Fixed in `get_recent_clinics()` by using `ExpressionAttributeNames` with `#cond` and `#lang` aliases. Without this fix, the Recent Clinics panel returns empty silently.

- **Delivery forms:** clinic_base_v1.txt updated to always include all available delivery forms (e.g. both KwikPen and single-dose vials for tirzepatide). Style file is read fresh on every pipeline run — no Lambda redeploy needed.

11 Quacklytics Portfolio

11.1 Portfolio Overview

Instrument	Status	Description
Illuminate	Live	Pharmaceutical intelligence infographics
Clinic	Live	Point-of-care patient education
Canvas	Scoped, not started	General-purpose visual intelligence
Resolve	Architected, not started	Healthcare entity identity resolution

11.2 Quacklytics Resolve

Healthcare entity identity resolution using NetworkX graph clustering and D-Wave quantum annealing. Unifies NPI, TIN, CCN, DEA, NDC, CMS LDS, IDN registries, and vendor IDs into a single canonical entity record with version history, confidence scoring, and audit-ready provenance.

Architecture: NetworkX graph (records as nodes, matches as weighted edges, connected components as canonical entities) + D-Wave hybrid solver for ambiguous partition optimization (QUBO, up to 2M variables, AWS Marketplace) + DynamoDB single-table entity registry + Amazon Neptune for relationship graph.

Patent FIR0001: Boltzmann machine training + quantum annealing for healthcare datasets.

Why quantum: With 8+ source systems and millions of records, valid partition assignments exceed 10^{56}. Classical exhaustive matching is intractable. Quantum annealing finds near-optimal solutions via quantum tunneling.

11.3 Strategic Positioning vs. MimiLabs

Forte pays $30K/year for MimiLabs — a Databricks wrapper around public CMS, CDC, and FDA datasets with natural language query.

	MimiLabs	Quacklytics
Data	Public only	Public + proprietary (IQVIA, LDS, Definitive)
Output	Query results / tabular	Original generated artifacts
AI role	NL to SQL	Synthesis, generation, identity resolution
Identity resolution	None	Quacklytics Resolve
Cost	$30,000/year	~$0.25 per generation

MimiLabs gives you ingredients. Quacklytics gives you the meal.

11.4 CMS DUA Compliance Note

CMS updated DUA renewal requirements effective August 11, 2026, requiring annual public dissemination of findings as a condition of extension. Forte licenses CMS LDS and must comply. Illuminate can produce compliant audience-ready documentation automatically. Every other LDS licensee faces the same obligation — a potential market beyond Forte.

12 Lambda Architecture — Current State

Function	Status	Job
illuminate-api-handler	LIVE ✅	GET /recent, POST /generate (async), background self-invoke, GET /status/{job_id}
clinic-api-handler	LIVE ✅	Same pattern. condition+drug+persona+language input. Multiple generations confirmed.

13 Cost Estimate (Per Generation)

Service	Standard	High	Ultra 4K
gpt-image-2	$0.19	$0.25	$0.40
Claude synthesis + prompt eng.	$0.05	$0.05	$0.05
Lambda, S3, DynamoDB, CloudFront	<$0.01	<$0.01	<$0.01
Total	**$0.24**	**$0.30**	**$0.45**

14 Backlog — Known Gaps & Future Enhancements

14.1 Recent Illuminations — Per-Resolution Tracking

IlluminateIndex uses drug_name + indication as key — one entry per drug regardless of resolution. Requires DynamoDB key schema migration. Deferred until after Ed/Darren review.

14.2 Therapeutic Area Style Library

Current Illuminate style library is oncology-biased. Proposed: S3 style layer files per TA (`style/oncology_v1.txt`, `style/metabolic_glp1_v1.txt`, etc.), selected by a new `therapeutic_area` key in the synthesis agent output. Style layers are also a commercial asset — configurable per client/indication.

14.3 Clinic — QR Code Implementation

The clinic_base_v1.txt style file instructs gpt-image-2 to include a QR code placeholder. Real QR codes linking to condition-specific patient resources are a future enhancement.

14.4 Clinic — Audience Mode Matrix Infographic

An infographic showing all 48 persona/language combinations (6 personas × 8 languages) with sample output differences is in the backlog for the conference deck.

15 Change Log

Version	Date	Change
3.1	2026-07-03	**Clinic subdomain live. DynamoDB reserved word bug fixed. Style file updated for delivery forms. Quick Reference section added.** clinic.quacklytics.org live: S3 bucket clinic-quacklytics-org created, static website hosting enabled, bucket policy added, index.html uploaded. New CloudFront distribution d3hu7iaz3hezk9 created (Free plan). Route 53 A alias record added for clinic.quacklytics.org. quacklytics-website distribution (d265he30e9rch9) upgraded to Pro plan ($15/month) to unlock 4th distribution slot. **Bug: DynamoDB reserved words.** `condition` and `language` are DynamoDB reserved words. The `get_recent_clinics()` scan was silently returning empty results. Fixed by adding `ExpressionAttributeNames` with `#cond` and `#lang` aliases in clinic api_handler.py. Lambda redeployed. Recent Clinics panel now populates correctly. **Style file update:** clinic_base_v1.txt updated to always include all available delivery forms (e.g. KwikPen and vials for tirzepatide). No Lambda redeploy needed — style file read fresh on every run. **Confirmed generations:** atrial fibrillation + rivaroxaban (Older Adult / Spanish, 177s) and obesity + tirzepatide (Tech Professional / English). **Quick Reference section added** with all critical identifiers, endpoints, distributions, buckets, tables, secrets, and deployment commands in one place.
3.0	2026-07-03	**Quacklytics Clinic built and first generation confirmed. URL architecture established. Portfolio expanded to four instruments.** Wildcard ACM cert *.quacklytics.org issued. illuminate.quacklytics.org live (new distribution duvaokvqrh5ct, new S3 bucket illuminate-quacklytics-org). Portfolio landing page rebuilt with four product cards. Route 53 AmazonRoute53FullAccess added to QuacklyticsAutoViz group. Clinic full pipeline built: api_handler.py, research_agent.py, synthesis_agent.py (6 personas, 8 languages, 15-key schema), prompt_engineer.py, storage_agent.py. DynamoDB ClinicIndex (PK: condition, SK: drug) and ClinicJobs created. S3 style file clinic_base_v1.txt uploaded. Lambda clinic-api-handler deployed. IAM self-invoke policy updated for both Lambda ARNs. API Gateway clinic-api deployed. First Clinic generation: type 2 diabetes + tirzepatide, general_adult, English, 3m12s. Resolve architecture documented (NetworkX + D-Wave, Patent FIR0001). MimiLabs positioning documented. CMS DUA August 2026 noted.

Version	Date	Change
2.5	2026-07-01	Bug fixes: research agent cache fallthrough, storage agent s3_source_keys preservation, resolution-aware cache. Root cause of Zepbound/Darzalex hallucination incident documented and all three bugs fixed. Drug list expanded. UI scrollable Recent Illuminations panel.
2.4	2026-07-01	Full async pipeline live end-to-end. Demo-ready. Four infographics confirmed: isatuximab, dupilumab, rasburicase, tumor lysis syndrome.
2.3	2026-06-30	Async backend. job_store.py. IlluminateJobs table. Self-invoke IAM policy.
2.2	2026-06-30	Async polling pattern diagram.
2.1	2026-06-30	Storage Agent, API Handler, API Gateway, UI overhaul.
2.0	2026-06-29	CMS Part D, Part B/D routing, Prompt Engineering Agent, E2E test. Resolution tiers.
1.9	2026-06-29	Authentication architecture. Two-site UI.
1.8	2026-06-29	Research Agent and Synthesis Agent complete.
1.7–1.0	2026-06-27–28	Initial build: DALL-E confirmed, sources, DuckDB, domain, frontend, Lambda architecture.

clinic_flow.png

Figure 3: Quacklytics Clinic pipeline architecture. Green accent palette distinguishes Clinic from Illuminate. Shows dual condition+drug source fetch, 6 persona matrix, 8 language support, and warm clinical style output. First confirmed generation: atrial fibrillation + rivaroxaban, Older Adult, Spanish, 177 seconds.